CS4074 – Big Data Analytics

Lab 1: Hadoop Installation & HDFS Operations

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1. Introduction

The objective of this lab was to install and configure Apache Hadoop in a single-node (pseudo-distributed) mode and to perform basic operations using the Hadoop Distributed File System (HDFS). The lab focused on setting up the required software environment, starting Hadoop services, interacting with HDFS using command-line tools, and analyzing file block distribution.

By completing this lab, I gained hands-on experience with Hadoop installation, HDFS management commands, and understanding how data is stored and replicated in a distributed file system.

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2. Hadoop Installation and Configuration

The lab began with installing OpenJDK 11, which is required for Hadoop to run. After verifying the Java installation, the JAVA_HOME environment variable was configured correctly.

Next, passwordless SSH was set up for localhost communication, which is essential for Hadoop services to start and communicate even in a single-node cluster.

Apache Hadoop version 3.3.6 was downloaded, extracted, and installed under /opt/hadoop. Environment variables such as HADOOP_HOME and PATH were configured to allow Hadoop commands to run from any terminal.

The Hadoop configuration files (core-site.xml and hdfs-site.xml) were modified to enable HDFS in pseudo-distributed mode with a replication factor of 1. Required directories for the NameNode and DataNode were created, and the NameNode was formatted successfully.

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3. Starting Hadoop and Verifying Services

After configuration, HDFS services were started using the start-dfs.sh command. The jps command confirmed that all required Hadoop processes were running:

- NameNode
- DataNode
- SecondaryNameNode

Additionally, the Hadoop NameNode Web UI was accessed through http://localhost:9870, confirming that the cluster was active and healthy.

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4. HDFS Operations

Several HDFS operations were performed during the lab:

- A user directory structure was created under /user/hams/.
- Files were uploaded to HDFS using the hdfs dfs -put command.
- Files were listed using hdfs dfs -ls.
- File contents were viewed using hdfs dfs -cat.
- Files were downloaded from HDFS to the local file system.
- A larger dataset (shakespeare.txt) was uploaded to analyze block distribution.

These steps demonstrated basic file management operations within HDFS.

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5. Block Distribution Analysis

The block distribution of shakespeare.txt was analyzed using the hdfs fsck command. The file system check showed that the file was stored correctly and that the HDFS cluster was in a healthy state.

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6. Issues Encountered and Solutions

During the lab, an SSH connection issue was encountered when connecting to localhost, resulting in a “broken pipe” message. This issue was resolved by restarting the SSH service and correcting file permissions for SSH keys. After fixing this, passwordless SSH worked correctly, allowing Hadoop services to start without issues.

No other major issues were encountered during the installation or configuration process.

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7. Answers to Required Questions

Question 1: How many blocks was shakespeare.txt split into?

The file shakespeare.txt was split into 1 block.

Question 2: What is the replication factor of your files?

The replication factor is 1.

Question 3: What is the default block size in your configuration?

The default HDFS block size is 128 MB.

However, since shakespeare.txt is smaller than the default block size, it occupies only a single block of approximately 5.4 MB.

Question 4: Why is the replication factor set to 1 in single-node mode?

The replication factor is set to 1 because the Hadoop cluster is running in single-node (pseudo-distributed) mode, where only one DataNode is available. Increasing the replication factor would not create additional replicas since there are no additional DataNodes.

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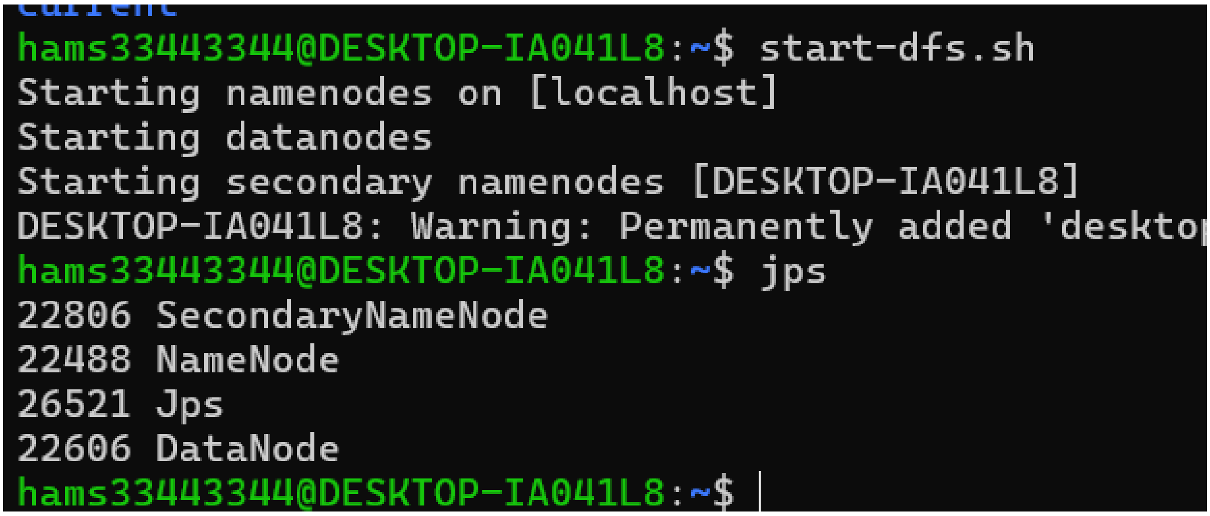
8. Conclusion

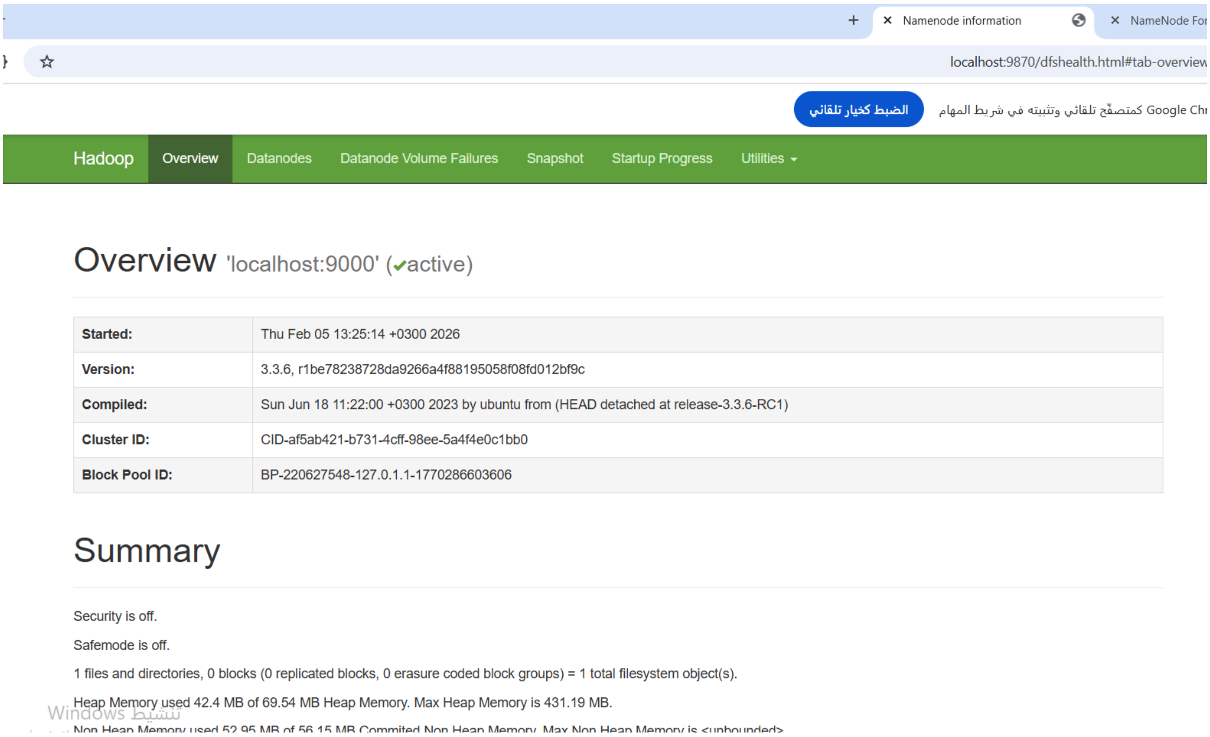
In this lab, Apache Hadoop was successfully installed and configured in pseudo-distributed mode. HDFS services were started and verified, files were uploaded and managed using HDFS commands, and block distribution was analyzed. This lab provided a strong foundation for understanding Hadoop architecture and prepared me for future labs involving MapReduce and large-scale data processing.

9. Screenshots

A:



B:



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Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).

Heap Memory used 50.08 MB of 69.54 MB Heap Memory. Max Heap Memory is 431.19 MB.

Non Heap Memory used 56.34 MB of 59.21 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	1006.85 GB
Configured Remote Capacity:	0 B
DFS Used:	28 KB (0%)
Non DFS Used:	4.47 GB
DFS Remaining:	951.16 GB (94.47%)
Block Pool Used:	28 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion (including replicas)	0

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NameNode Journal Status

Current transaction ID: 3

Journal Manager	State
FileJournalManager(root=/opt/hadoop/hdfs/namenode)	EditLogFileOutputStream(/opt/hadoop/hdfs/namenode/current/edits_inprogress_0000000000000000003)

NameNode Storage

Storage Directory	Type	State
/opt/hadoop/hdfs/namenode	IMAGE_AND_EDITS	Active

DFS Storage Types

Storage Type	Configured Capacity	Capacity Used	Capacity Remaining	Block Pool Used	Nodes In Service
DISK	1006.85 GB	28 KB (0%)	951.17 GB (94.47%)	28 KB	1

Hadoop, 2023.

C:

```
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -ls /user/hams/input/
Found 2 items
-rw-r--r--   1 hams33443344 supergroup         99 2026-02-05 15:21 /user/hams/input/hello.txt
-rw-r--r--   1 hams33443344 supergroup    5422721 2026-02-05 15:36 /user/hams/input/shakespeare.txt
```

D:

```
hams33443344@DESKTOP-IA041L8:~$ hdfs fsck /user/hams/input/shakespeare.txt -files -blocks
Connecting to namenode via http://localhost:9870/fsck?ugi=hams33443344&files=1&blocks=1&path=%2Fuser%2Fhams%2Finp
akespeare.txt
FSCK started by hams33443344 (auth:SIMPLE) from /127.0.0.1 for path /user/hams/input/shakespeare.txt at Thu Feb 0
:29 AST 2026
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```
/user/hams/input/shakespeare.txt 5422721 bytes, replicated: replication=1, 1 block(s):  OK
0. BP-220627548-127.0.1.1-1770286603606:blk_1073741826_1002 len=5422721 Live_repl=1


Status: HEALTHY
 Number of data-nodes:  1
 Number of racks:               1
 Total dirs:                    0
 Total symlinks:                0

Replicated Blocks:
 Total size:    5422721 B
 Total files:   1
 Total blocks (validated):      1 (avg. block size 5422721 B)
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Total blocks (validated):      1 (avg. block size 5422721 B)
Minimally replicated blocks:   1 (100.0 %)
Over-replicated blocks:        0 (0.0 %)
Under-replicated blocks:       0 (0.0 %)
Mis-replicated blocks:         0 (0.0 %)
Default replication factor:    1
Average block replication:     1.0
Missing blocks:                0
Corrupt blocks:                0
Missing replicas:              0 (0.0 %)
Blocks queued for replication: 0

rasure Coded Block Groups:
Total size:    0 B
Total files:   0
Total block groups (validated):        0
Minimally erasure-coded block groups:  0
Over-erasure-coded block groups:       0
Under-erasure-coded block groups:      0
Unsatisfactory placement block groups: 0
Average block group size:      0.0
Missing block groups:          0
Corrupt block groups:          0
Missing internal blocks:       0
Blocks queued for replication: 0
SCK ended at Thu Feb 05 19:14:31 AST 2026 in 1640 milliseconds


he filesystem under path '/user/hams/input/shakespeare.txt' is HEALTHY
ams33443344@DESKTOP-IA041L8:~$
```